

Comparison of VaR Methods: The Case of Indian Equities

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Abstract

Different approaches to calculate VaR are based on different assumptions. It is thus important to identify the most appropriate method of calculating VaR for different assets and markets. This paper deals with a comparative evaluation of four Value-at-Risk models namely Historical VaR, Normal VaR, GARCH (1,1) VaR and Volatility Weighted Historical Simulation (VWHS) VaR in terms of their prediction accuracy for an active portfolio of Indian equities. Daily NAVs of 34 Indian equity growth mutual fund schemes for a period of 10 years starting 27th November 2003 till 28th November 2013 are used to calculate daily 95% VaR and backtest the results using Kupiec's Test for all four VaR models. We conclude that there is a significant difference in the performance of different VaR models if there is a difference in the market conditions (reflected by highly contrasting returns of the underlying asset) during VaR calculation and VaR backtesting period. On the other hand, if the market conditions (and thus the returns of the underlying asset) are similar during the VaR calculation and VaR backtesting period, all VaR methods seem to perform equally well. In our study, VWHS VaR has performed exceedingly well in both scenarios. Hence, the study finds VWHS VaR to be a better methodology for measuring VaR of an active portfolio of Indian equities. The results are relevant for traders, retail and institutional investors who hold stocks of Indian companies in their portfolio and need to calculate VaR as a measure of market risk for their positions.

Key Words: *Value at Risk, Backtesting, Historical VaR, Volatility Weighted Historical Simulation VaR, Normal VaR, GARCH (1,1) VaR*

JEL Classification Codes: C52; C53; G21

1. Introduction

Over the last few decades financial markets have gone through a swift development and a large number of complex securities were introduced. This expansion resulted in a complicated environment where traditional risk measures proved to be insufficient and new measures were introduced. Two basic properties of financial markets are risk and return. While the calculation of rate of return is quite straightforward, capturing risk is more complex. The basic purpose of this study is to examine the accuracy of one of the most widely used and accepted risk measurement tools called 'Value at Risk' (VaR). To begin with, we shall discuss the backdrop of risk management and its evolution followed by an explanation of VaR and Backtesting.

Financial risk can be further sub-divided into market risk, liquidity risk, credit risk and operational risk. The focus of this study is market risk. Market risk is the risk of losses owing to movements in the level or volatility of market prices. It can be of two types: absolute risk, measured in currency terms and relative risk, measured relative to a benchmark index. While the former focuses on the volatility of total returns, the latter measures risk in terms of tracking error or deviation from the index. Market risk is controlled by limits on notional exposures, VaR measures, and independent supervision by risk managers.

1.1 Value at Risk

During the past two decades, VaR has become one of the most popular risk measurement techniques in finance. VaR is a method which aims to capture the market risk of a portfolio of assets. Put formally, VaR measures the maximum loss in value of a portfolio over a predetermined time period for a given confidence interval. As per (Holton, 2014), to specify a value-at-risk metric, we must identify three things:

- i. The period of time over which a possible loss will be calculated - 1 day, 2 weeks, 1 month, etc. This is called the value-at-risk horizon. In our study, the value-at-risk horizon is one trading day.
- ii. A quantile of that possible loss. For example, the portfolio's value-at-risk is usually expressed as a 95% or 99% quantile of loss.
- iii. The currency in which the possible loss is denominated. This is called the base currency.

A loss which exceeds the VaR threshold is termed a "Violation" or "Exception" or "Exceedance".

1.2 Backtesting

Despite the wide use and common acceptance of VaR as a risk management tool, the method has frequently been criticized for being incapable to produce reliable risk estimates. When implementing VaR models, there will always be numerous simplifications and assumptions involved. Moreover, every VaR model attempts to forecast future asset prices using historical market data which does not necessarily reflect the market environment in the future. Thus, VaR models are useful only if they predict future risks accurately. In order to verify that the results acquired from VaR calculations are consistent and reliable, the models should always be backtested with appropriate statistical methods. Backtesting is a procedure where actual profits and losses are compared to projected VaR estimates. (Jorion, 2001) refers to these tests aptly as 'reality checks'. If the VaR estimates are not accurate, the models should be reexamined for incorrect assumptions, wrong parameters or inaccurate modeling. Selection of an appropriate VaR methodology to capture market risk accurately is of prime importance with regard to relevance and utility of VaR as a risk measurement and management tool.

A variety of different testing methods have been proposed for backtesting purposes. Basic tests, such as Kupiec's (1995) Proportion of Failure (POF) test, examine the frequency of losses in excess of VaR. The failure rate should be in line with the selected confidence level. For instance, if daily VaR estimates are computed at 99% confidence for one year (250 trading days), we would expect on average 2.5 VaR violations to occur during this period. In the POF test we would then examine whether the observed amount of exceptions is reasonable compared to the expected amount. This is the technique of backtesting that is used in this study to compare VaR models.

In the next section, we review the existing literature on VaR methodologies and backtesting. In the third section, we discuss the research gap and motivation for the study. In the fourth section, we list the objectives and hypotheses. Fifth and sixth sections describe the data and methodology

used in the study respectively. Seventh section consists of the analysis and interpretation of the study whereas eighth section concludes the paper and provides relevant managerial implications.

2. Literature Review

The specification of return distribution and volatility is central to a VaR model as the accuracy of VaR estimates directly depends on the assumptions made with regard to these two crucial inputs in the VaR calculation. The extensive literature on VaR models and backtesting varies broadly on the grounds of these assumptions. (Linsmeier & Pearson, 1996) provide introduction to the concept of VaR and discuss basic methodologies. Plenty of studies deal with the question of the distribution that approximates empirical returns. First posed by Louis Bachelier in the turn of the 20th century, the Gaussian distribution is used for returns ever since. (Angelidis & Benos, 2006), (Berkowitz & O'Brien, 2002), and many others employ normal distribution in the models as its use is supported by the Central Limit Theorem. But the suitability of normal distribution for real financial data is still under constant discussion. (Frain, 2008) considers the fit of the actual (historical) data with normal distribution to be unsatisfactory, even misleading.

Value at Risk by (Jorion, 2007) covers the introduction to the concept of VaR and overall picture of market risk measurement. (Christoffersen, 1998), (Christoffersen & Diebold, 2000), and (Christoffersen & Pelletier, 2004) in their work employ conditional coverage to backtest VaR models whereas (Lopez, 1998) introduces the loss function. (Basel Committee, 1996) implemented VaR approach for regulatory purpose and introduced fundamentals of backtesting. From a wide range of works that cover overall picture on VaR developments, summarize its properties, individual methods with their strengths and weaknesses, cover the critique towards individual VaR models, and offer alternative approaches to risk management, we draw inspiration mainly from (Jorion, 2007) and (Dowd, 1998).

According to (Nolan, 2003), stable distribution is preferred by some researchers as it is considered a better fit for tail distribution of actual returns. Nolan shows that return distributions with heavy tails are well described by stable distributions, or at least much better than by Gaussian models. On the other hand, (Barunik et al., 2010) confirmed that real data has fatter than normal but lighter than stable tails and the use of stable distribution causes overestimation

of extreme returns. (Frain, 2008) confirms this property on models that contain stable distribution. (Giot & Laurent, 2004) compared several volatility models: EWMA, an asymmetric GARCH and Residual Volatility. The models are estimated with the assumption that returns follow either normal or skewed t-Student distributions. They found that under a normal distribution, the Residual Volatility model performed best. However, under a skewed t-distribution, the asymmetric GARCH and Residual Volatility models provided very similar results. The authors reported that the superiority of the models based on Residual Volatility over the GARCH family is not as obvious when the estimation of the latter assumes the existence of asymmetric and leptokurtic distributions. (Níguez, 2008) compared the ability to forecast VaR using different GARCH family models (GARCH (1,1), AGARCH, APARCH, FIGARCH and FIAPARCH, and EWMA) and found that the combination of asymmetric models with fractional integrated models provided the best results. Although this evidence is somewhat ambiguous, the asymmetric GARCH models seem to provide better VaR estimations than the symmetric GARCH models.

3. Research Gap and Motivation

For financial institutions, the possibility of sudden losses is unwanted and important to track. Therefore, VaR methodology employs the lower quantiles, which measures only the possibility of an exceptional loss. The strength of VaR measurements lies in the combination of forward looking (ex-ante) risk control, ease of explanation and aggregation of risks. The concept of VaR is quite easy to understand for a wide audience since the overall quantity of potential losses is aggregated into one value incorporating risk from interest rates, equities, commodities and foreign currencies. The pitfall of VaR lies in the necessity to choose a computational method for volatility and a suitable distribution of returns. Choosing an appropriate VaR method is an important and difficult task, and risk managers have coined the term Model Risk to cover the hazards of working with potentially mis-specified models. If a used model underestimates risk, bank's capital buffer might be insufficient to take up the loss or financial regulator may impose penalties. On the other hand, excessive VaR implies an ineffective use of capital. Therefore, it is essential to find a method which accurately specifies the VaR estimate.

The concept of Value at Risk (VaR) has quickly become the standard for measurement of market risk in the financial sector for both financial institutions and financial regulators (Engle & Manganelli, 1999) for its ease of use and clarity of meaning. Systems for internal risk measurement progress in parallel with a regulatory framework. Basel II framework identifies VaR analysis as a preferred tool for assessing exposure of a bank to market risk. As per the Basel report titled “Revisions to the Basel II market risk framework” released in February 2011, following are some key points relevant to our discussion.

- No particular type of model is prescribed. So long as each model used captures all the material risks run by the bank, they will have flexibility in devising the precise nature of their models, but the following minimum standards will apply for the purpose of calculating their capital charge. Individual banks or their supervisory authorities will have discretion to apply stricter standards.
 - a. “Value-at-risk” must be computed on a daily basis.
 - b. In calculating the value-at-risk, a 99th percentile, one-tailed confidence interval is to be used.
 - c. The choice of historical observation period (sample period) for calculating value-at-risk will be constrained to a minimum length of one year. For banks that use a weighting scheme or other methods for the historical observation period, the “effective” observation period must be at least one year.
 - d. Each bank must meet, on a daily basis, a capital requirement expressed as the sum of: The higher of either its previous day’s value-at-risk number or an average of the daily value-at risk measures on each of the preceding sixty business days (VaR_{avg}), multiplied by a multiplication factor (m_c). The multiplication factors (m_c) will be set by individual supervisory authorities on the basis of their assessment of the quality of the bank’s risk management system, subject to an absolute minimum of 3.

Given that banks are permitted to develop differing VaR models, it is necessary for supervisory bodies to be able to assess the relative performance of these alternative models.

VaR is multi-purpose tool that can be used by individual investors, traders, institutional investors, banks etc. to assess the underlying risk of incurring large losses on their asset

portfolio. The importance of accurately calculating VaR from a regulatory and compliance point of view is also paramount as discussed above.

4. Objectives and Research Hypotheses

We use data of 34 Indian equity (growth) mutual fund schemes to calculate VaR and perform backtesting over a period of 10 years. Thus, the results of this study may well be generalized for Indian equity markets. Another prominent aspect of this study is the use of actively managed portfolios rather than passive market indices. This is in line with the practical reality whereby banks have to calculate VaR for their actively traded asset portfolios. Hence we expect to obtain results of practical relevance in this regard. The identification of suitable models to calculate VaR for Indian equity portfolios is useful for traders and investors who deal in Indian equities.

Following are the objectives of this study.

- i. To evaluate performance of different approaches and methods of VaR estimation by backtesting for a portfolio of Indian equities. Put simply, the objective of this study is to determine the accuracy of VaR models.**
- ii. To verify whether the selection of similar or contrasting data periods in terms of market conditions for VaR calculation and VaR backtesting leads to a difference in the performance of VaR models.**

The research hypothesis of the study is that VaR models differ in terms of their accuracy of VaR estimation and that this accuracy depends on the extent of congruence in the market conditions (and thus the behavior of returns of underlying asset) during VaR calculation and VaR backtesting periods.

5. Data

The data for this study has been taken from Capitaline database for 34 equity growth schemes pertaining to 20 asset management companies (AMCs) operating in India. Daily Net Asset Values (NAVs) for all schemes were collected for a period of 10 years starting 28th November 2003 till 27th November 2013. To begin with we were able to find data for a total of 330 equity growth schemes for this ten year period. With a standard norm of approximately 250 trading days in a year, a fund should have 2500 NAVs for a period of 10 years. Due to missing values in the data none of the 330 schemes had 2500 NAVs for the 10 year period under consideration. By allowing a maximum limit of 5% missing data, we decided to rule out those schemes from our analysis which had less than 2375 data points. This reduced the number of schemes to 54. The next step was to deal with the dates for which majority of the schemes did not have an NAV. For instance, these dates were either a Saturday or another non-trading day for which majority of the schemes did not have an NAV. We decided to remove these dates from the analysis. This is because imputation of data for a large number of days and for majority of schemes might lead to misleading results. We came down to a figure of 2315 days after removing the dates for which a majority of schemes did not have a data point. We found out that 17 schemes had data for all 2315 days. To increase the number of schemes under analysis, we decided to include all schemes which had 2250 or more data points. This effectively corresponds to setting a limit of at least 90% availability of data (2250 is 90% of 2500) to shortlist the schemes for analysis. This restriction reduced the number of schemes from 54 to 45. The next step was to select schemes eligible for data imputation as these 45 schemes still had missing net asset values for some dates. For this purpose, we decided to rule out schemes which had missing NAVs for a minimum of 2 consecutive days. This meant that we included only those schemes in the analysis which did not have a single instance of consecutive missing values. This was done because it was not feasible to impute values for a consecutive set of missing dates for a particular scheme. This step further reduced the number of schemes for analysis to 34 out of which 17 schemes had data for all 2315 days i.e. 95% data was available (2315 is 95% of 2500) for these schemes. For the other set of 17 schemes which had missing values, the missing data points were imputed by taking an average of previous and next day's values. Finally we had 34 equity growth schemes spread across 20 asset management companies available for the analysis. This set of 34 actively managed equity portfolios is taken as a representative of investments made in the Indian equity

market by retail and institutional investors. We believe that the results of VaR backtesting on this set of 34 equity schemes can be a guiding force to identify the most accurate VaR methodology for a portfolio comprising of Indian equities.

6. Methodology

VaR is a forward looking measure of market risk in the sense that it tries to estimate the maximum potential loss that can be incurred given a confidence level and time horizon. This is one of the main reasons of its wide popularity and acceptance. But the irony is that VaR is calculated using historical data. This implies that if VaR is calculated in a relatively tranquil period with positive returns on an average, it shall produce an estimate which is not suitable for a crisis situation having large negative returns and vice-versa. This indicates that VaR is heavily dependent on historical data and assumes that the returns in near future shall be consistent with the returns in the past. Modifications can surely be made to VaR methods to overcome this issue. In fact methods which dynamically adjust VaR estimates for volatility changes can produce better results by predicting future returns. Keeping this in view, we decided to divide our 10 years of daily NAV data into two parts: Pre-Crisis Evaluation and Post-Crisis Evaluation. Our total data starting in 2003 and ending in 2013 has a phase of severely distressed returns during the period of financial crisis of 2008-2009. This global meltdown owing to the sub-prime mortgage crisis in the US had its fair share of impact on equity returns in India. According to reference dates of business cycles issued by National Bureau of Economic Research (NBER, USA), the US business cycle reached its peak in December 2007 and trough in June 2009. To mark the crisis period accordingly, we have taken a period of two years starting 1st January 2008 till 31st December 2009 to represent the phase of financial distress. The trough might have been reached in June 2009 but since the economy takes time to recover and equity returns stabilize gradually, we decided to add another 6 months and stretch the period of distress to 31st December 2009 making it a two-calendar year long phase. VaR backtesting has been applied by adopting the intuitive out-of-sample approach. This means that VaR is first calculated over a period of time and then backtesting is applied ahead of this time to test whether VaR estimates are able to stand the test of time by predicting frequency of violations accurately or not. This out-of-sample approach is appropriate to compare VaR methods because it examines the relevance of

a VaR estimate as a forward looking risk measure. It helps us to identify which methods are able to produce future-oriented estimates on the basis of historical data. If a method sustains the out-of-sample backtesting approach and produces lesser number of violations than the level of significance, it is an appropriate method of calculating VaR. Now for the first part of analysis i.e. Pre-Crisis Evaluation, we calculate VaR over a four-year period i.e. 27th November 2003 till 31st December 2007 and backtest the estimates over a two-year period of financial crisis i.e. 1st January 2008 to 31st December 2009. This implies that we try to find out which VaR methods produce accurate estimates of VaR (on the basis of historical data during period of relatively stable returns) which are able to withstand backtesting during the turbulent phase of financial crisis spanning two calendar years. For the second part of analysis i.e. Post-Crisis Evaluation, we calculate VaR over a two-year period i.e. 1st January 2010 till 31st December 2011 and backtest the estimates over a 23-month period i.e. 1st January 2012 to 28th November 2013. This implies that we try to find out if there is a difference between quality of VaR estimates produced by different VaR methods if the VaR calculation period and backtesting period are largely similar in terms of economic conditions.

After dividing the data into relevant periods for a two-fold analysis, we calculate VaR by using four methods that collectively represent all three approaches of VaR computation: Parametric, Non Parametric and Semi Parametric.

6.1 Parametric VaR Methods

- a. **Normal VaR:** VaR is calculated with the assumption that returns follow a normal probability distribution. The volatility is calculated as standard deviation of the return series and plugged in the formula for VaR. The one tailed standard normal variate corresponding to the chosen confidence level is multiplied by the volatility to calculate the extent of deviation from the mean to arrive at VaR figure.

$$VaR(\alpha) = F^{-1}(\alpha) = \mu_t + \sigma_t(\alpha)$$

Where, μ represents mean, σ represents historical standard deviation, α denotes confidence level and t denotes time period considered.

- b. **GARCH (1,1) VaR:** The only difference between Normal VaR and GARCH (1,1) VaR is the difference in calculation of volatility to be inserted in the VaR calculation. This methodology

produces a forward looking estimate of volatility and hence can predict VaR more accurately if there is volatility clustering in the return series. In this method, σ is calculated using Generalised Auto-Regressive Conditional Heteroskedasticity (GARCH (1,1)). GARCH (1,1) is estimated using following equations:

$$R_t = \mu_t + \varepsilon_t$$

$$\sigma_t^2 = \omega + \sum_{j=1}^q \alpha_j [\varepsilon_{t-j}^2] + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

where, R_t represents return, μ_t represents mean return and ε_t is error term in mean equation; σ_t is volatility, coefficient ω is constant, α_j represents ARCH term and β_j represents GARCH term.

6.2 Non-Parametric VaR Method

a. Historical VaR

In this method, VaR is estimated without taking into consideration any assumption about the underlying return distribution. It is based on the assumption that the behavior of returns in the past shall be the same in future. It is the simplest method of estimating VaR. The justification for this methodology is that if returns are stationary then empirical distribution is a consistent estimator of the unobserved future distribution function. This method is defined as follows. Consider a sample of past ω returns. The historical VaR at α level of significance is given by:

$$VaR_\alpha = -Q_{1-\alpha}(r_t, r_{t-1}, \dots, r_{t-\omega+1})$$

where, r_t is return of the asset under consideration at time t and Q is the relevant quantile function at α level of significance.

6.3 Semi-Parametric Methods: The semi-parametric methods combine the traditional simulation model with conditional volatility models (like GARCH (1,1)), which makes this technique attractive in dealing with volatility fluctuations.

a. Volatility Weighted Historical Simulation

This method was proposed by (Hull & White, 1998). It combines the benefits of historical simulation with volatility updating. The basic premise of this approach is to update return information with recent changes in volatility. According to this approach, $\text{VaR}(\alpha)$ is the α quantile of the distribution of the volatility adjusted return. In this methodology all past returns are scaled by volatility adjustment factor. Volatility adjusted returns are as follows.

$$R_{t,i} = r_{t,i} * \frac{\sigma_{T,i}}{\sigma_{t,i}}$$

Where, $r_{t,i}$ is actual return for asset i on day t ; $\sigma_{T,i}$ is current forecast of volatility for asset i ; $\sigma_{t,i}$ is volatility forecast for asset i on day t (made at the end of day $t-1$).

6.4 Backtesting of VaR

The most common backtesting method for a VaR model is to count the number of VaR exceptions, i.e. days (or holding periods of other length) when portfolio losses exceed VaR estimates. This method is known as the Kupiec Test as it was suggested by Paul Kupiec in 1995. If the number of exceptions is less than the selected level of significance, the VaR model overestimates risk. On the contrary, too many exceptions signal underestimation of risk. Naturally, it is rarely the case that we observe the exact amount of exceptions as suggested by the confidence level. It therefore comes down to statistical analysis to decide whether the number of exceptions is reasonable or not, i.e. will the model be accepted or rejected.

Denoting the number of exceptions as x and the total number of observations as T , we may define the failure rate as x/T . In an ideal situation, this rate would reflect the selected level of significance. For instance, if a confidence level of 99 % is used, we have a null hypothesis that the frequency of tail losses is equal to $p = (1 - c) = 1 - .99 = 1\%$. Assuming that the model is accurate, the observed failure rate x/T should act as an unbiased measure of p , and thus converge to 1% as sample size is increased. (Jorion, 2001)

Each trading outcome either produces a VaR violation or not. This sequence of ‘successes and failures’ is commonly known as Bernoulli trial. The number of exceptions x follows a binomial probability distribution:

$$f(x) = p^x(1 - p)^{T-x}$$

As the number of observations increase, the binomial distribution can be approximated with a normal distribution:

$$z = \frac{x - pT}{\sqrt{p(1 - p)T}} \approx N(0,1)$$

where pT is the expected number of exceptions and $p(1 - p)T$ is the variance of exceptions. (Jorion, 2001)

By utilizing this binomial distribution we can examine the accuracy of the VaR model. However, when conducting a statistical backtest that either accepts or rejects a null hypothesis (of the model being ‘good’), there is a tradeoff between two types of errors. Type 1 error refers to the possibility of rejecting a correct model and type 2 error refers to the possibility of not rejecting an incorrect model. A statistically powerful test would efficiently minimize both of these probabilities. (Jorion, 2001)

Hence, the only information required to implement a Kupiec’s Proportion of Failure (POF) test is the number of observations (T), number of exceptions (x) and the confidence level (c). (Dowd, 2006)

7. Analysis and Interpretation

We have divided our analysis into separate two heads viz. Pre Crisis Evaluation and Post Crisis Evaluation. We shall start by studying the results of Pre Crisis Evaluation. The key phenomenon to keep in consideration is that for Pre Crisis Evaluation, we calculate VaR over a four-year period i.e. 27th November 2003 till 31st December 2007 and backtest the estimates over a two year period of financial crisis i.e. 1st January 2008 to 31st December 2009. Following is a table showing results of the Pre-Crisis Evaluation:

Table 1: Pre Crisis Evaluation: Summary of Backtesting Results

Category of VaR Method	Non Parametric	Parametric		Semi Parametric
Null Hypothesis: $x/T = \text{Significance Level}$	Historical VaR	Normal VaR	GARCH (1,1) VaR	VWHS VaR
No. of Hypotheses 'Not Rejected' out of 34	13	2	1	24
Success Ratio	38.24%	5.88%	2.94%	70.59%

The null hypothesis being tested in the Kupiec test is that proportion of failure i.e. x/T (x is the number of violations and T is the total number of observations) is equal to α i.e. level of significance. If this null hypothesis is not rejected, the VaR method passes the Kupiec Test. This is because the total number of violations observed in the testing period is not statistically different from the level of significance. The level of significance has been fixed at 5% for all hypotheses tests conducted in this study. For each VaR method, the success ratio has been defined as ratio of number of schemes for which the null hypotheses is not rejected and total number of schemes i.e. 34.

As can be observed from the above table, out of the total 34 schemes for which VaR has been calculated and backtested, VWHS VaR has been able to accurately predict VaR figures for all 24 schemes i.e. the success ratio of VWHS VaR is 70.59%. This means that for 24 schemes, there is no difference between the proportion of failure observed in the backtesting period and the significance level of 5%. This shows that standardizing the daily log returns by GARCH volatility estimates tweaks a return series in such a way that it gives accurate VaR estimates calculated as the relevant quantile of the distribution.

The next most accurate model is Historical VaR with a success ratio of 38.24%. This shows that for 13 schemes, for which the null hypotheses has not been rejected, the VaR calculation period had returns low enough to match the distressed returns observed in the backtesting period. The VaR limit was set in such a way that it could not bring a difference in the number of violations and level of significance. Parametric VaR models i.e. Normal VaR and GARCH (1,1) VaR have performed poorly in the backtest with a success ratio of 5.88% and 2.94% respectively. This is

mainly due to the use of standard normal variate to determine VaR quantile whereas the return distribution is far from normal.

We now move on to the analysis of results of Post-Crisis Evaluation. In this case, we calculate VaR for a two year period i.e. 1st January 2010 till 31st December 2011 and backtest the estimates over a 23-month period i.e. 1st January 2012 to 28th November 2013. The period of VaR calculation and VaR backtesting are broadly similar in the sense that none of the periods had experienced a situation of highly distressed or superlative returns. It is interesting to know whether different VaR methods perform similarly if there is no significant difference between the VaR calculation and VaR backtesting period. Following is a table showing results of the Post-Crisis Evaluation.

Table 2: Post Crisis Evaluation: Summary of Backtesting Results

Category of VaR Method	Non Parametric	Parametric		Semi Parametric
		Normal VaR	GARCH (1,1) VaR	
Null Hypothesis: $\alpha/T = \text{Significance Level}$	Historical VaR			VWHS VaR
No. of Hypotheses 'Not Rejected' out of 34	34	29	34	33
Success Ratio	100.00%	85.29%	100.00%	97.06%

We observe that for post crisis evaluation, Historical VaR and GARCH (1,1) VaR produce a success ratio of 100%. In fact, Normal VaR and VWHS VaR also give highly accurate results with a success ratio of 85.29% and 97.06% respectively. Broadly all methods perform well in this scenario where the data used for VaR calculation is quite similar in its behaviour to the data used for VaR backtesting. The superior performance of Historical VaR shows that the VaR estimates obtained from the VaR calculation period were accurate as the proportion of violations in the backtesting period were not different from the level of significance. The same is true for GARCH (1,1) VaR and VWHS VaR. However, Normal VaR did not perform equally well as the normality assumption for returns did not hold true for this analysis either. It is because of this similarity in results that our first analysis of pre-crisis evaluation becomes even more important as it is difficult to differentiate among the relative performance of VaR methodologies in the post-crisis evaluation. However, VWHS has emerged as a clear winner in both parts of the

analysis. It has accurately predicted VaR estimates in pre-crisis and post-crisis evaluation delivering sound performance each time. Hence, we conclude that Volatility Weighted Historical Simulation is the best methodology out of the four different methods compared in this study for measuring VaR for an active portfolio of Indian equities.

8. Conclusion and Managerial Implications

Value at Risk is a widely used measure of risk today because of its simplicity and flexibility. The variants of VaR are evident its suitability and universal appeal as a standard measure of risk. It is a measure that needs to be reported by commercial banks as a regulatory compliance according to the Basel norms. But the flexibility of VaR is also one of its limitations. Different approaches to calculate VaR are based on different assumptions. These assumptions relate to the underlying distribution of returns and measurement of volatility. It is thus important to identify the most appropriate method of calculating VaR for different assets and markets. In this study we tried to identify the most suitable VaR methodology for an active portfolio of Indian equities. To get a broad representation of Indian equities, we performed Kupiec's test of backtesting on 34 equity growth mutual fund schemes over a two-fold period separated by the financial crisis of 2008-09. The VaR methodologies considered were Historical VaR, Normal VaR, GARCH (1,1) VaR and Volatility Weighted Historical Simulation (VWHS) VaR. We conclude that there is a noticeable difference in the performance of different VaR models if there is a difference in the market conditions as reflected by highly contrasting behavior of return series in the VaR calculation and VaR backtesting period. On the other hand, if the market conditions and thus the behavior of returns are similar during the VaR calculation and VaR backtesting period, all VaR methods seem to perform equally well. In our study, VWHS VaR has performed exceedingly well in both scenarios. Hence, we conclude that VWHS VaR is the best methodology out of the four different methods compared in this study for measuring VaR of an active portfolio of Indian equities. The shortcoming of this study is the limited number of VaR methods considered to measure market risk of a portfolio consisting of Indian equities. However, the study points out at the relative precision of semi-parametric methods in comparison to parametric and non-parametric techniques due to incorporation of volatility fluctuations and non-normality of return distribution. The results are relevant for traders, retail and institutional investors who hold stocks

of Indian companies in their portfolio and need to calculate VaR as a measure of market risk for their positions.

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